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ORIGINAL ARTICLE

Artificial intelligence-based smart galliformes farm management system

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Abstract

This article introduces an Artificial Intelligent-driven system for Galliformes Farm Management, consulting, and disease control. Comprising both a web-based mobile app and a website, the system integrates physical electronic devices to regulate intelligent automation. The application employs Artificial Neural Network and computer vision for the identification of breeds and the recognition of diseases in galliform birds, utilizing TensorFlow for real-time detection. It notifies local farmers about bird flu infections with an integrated infection map using Google's map Application Programming Interface. The system also controls farm temperature and humidity through microcontrollers and sensors, offering consulting features such as cost calculation, food chart generation, and drug suggestions.

Keywords Poultry farm automation · Artificial intelligence · Breed identification · Disease classification · Smart farm management · Image classification

1 Introduction

During COVID-19, we have witnessed various modern changes around us. As a result, digitalization has profoundly influenced our daily lives, the job market, skills demand [1], healthcare, education, the economy, society, agriculture, and various industrial sectors. This change has indicated to us that smart systems based on the Internet and the adaption of various cloud technologies [2] can assist us in various ways. During the pandemic, attention shifted from corporate jobs to businesses and agriculture. Starting businesses in third-world countries is difficult due to high costs, but poultry farming requires less capital and significant experience. A smart AI-based app for identifying hen breeds, diseases, and farm management could help novice farmers. Throughout the years, some research has been carried out, and at the same time some software [3–6] and IoT-based systems [7–10] have been partially developed, which increased poultry production.

In modern poultry farming, smart technologies revolutionize management practices. Using sensors, IoT devices and data analytics, farmers monitor parameters in real time, optimizing conditions, and ensuring bird welfare. Automated feed and water systems minimize waste and improve feeding consistency. Advanced imaging aids in the early detection of disease, preventing outbreaks. Smart environmental controls maintain optimal conditions for growth and productivity. Remote monitoring and cloud-based platforms enable real-time farm management from anywhere. Integration with precision farming techniques optimizes resource utilization and sustainability. In general, these innovations drive efficiency, sustainability, and profitability in poultry farming, marking a transformative change in industry practices. In existing

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solutions, many of the applications are common to our app, except the following three: breed identification, disease recognition, and bird flu notification on the map.

This smart mobile application aims to manage poultry farms and provide consultancy. Using the app, users will be able to identify bird breeds. This kind of breed identification has been seen as a challenge and researchers have been conducting extensive experiments to address the issue in sheep breed classification [11], cattle classification [12], and similar fields. This app also aims to classify diseases and, at the same time, disease infection notifications will be sent to users. The app will also be able to smartly control the firm from a distance. Users can also receive medical assistance from the app.

The 'Galliformes Farm Management System' [13] is an intelligent automated system that will help a person manage a poultry farm with helpful advice regarding poultry farming without prior knowledge. This system will also help to control diseases and alert local farmers about them. Also, artificial intelligence is used to recognize bird breeds and identify specific diseases with the help of artificial deep neural networks, which is a popular machine learning Algorithm for image recognition. For the classification of hen breeds, the model was trained on the 'GalliformeSpectra' data set [14]. In addition to these, this system will also provide an efficient cost calculation regarding agriculture,farming, food nutrition, and medicine consulting in case of diseases. This system will use Google Maps to pinpoint the bird flu locations to the user so that local people can be aware of that and take the necessary measurements. Along with these core features, there are related necessary features like statistical analysis using web data scraping technology, blog, live chat, donation system, etc. The Galliformes Farm Management application system integrates multiple user-centric modules to enhance farm management. Its User-centric interfaces [15] handles registration, login, and profile adjustments, while the bird breed identification and disease recognition system utilize machine learning for breed and disease detection, including real-time analysis via cameras. The farming cost calculation and food chart generation modules offer online functionality for cost management and personalized bird food charts. The system also includes a medicine suggestion module, bird flu reporting with google maps integration, and automation for remote farm monitoring. The live chat module enables communication with authorities and veterinarians during emergencies. Additional features include donation support, a blog platform, and business analysis reports driven by machine learning. Overall, the system aims to improve efficiency, communication, and data-driven decision-making in poultry farming, offering robust functionalities for both guest and authenticated users. Table 1 represents some related app products.

We have reviewed some of the world's leading poultry farm management and automation systems. Common features of poultry automation systems are given below.

- Personalized user profile.
- Generation of food charts.
- Poultry automation using smart devices.
- Inventory management systems.
- Humidity & and temperature update.

Limitations of current automatic poultry management systems:

- Absence of automatic breed identification using machine learning feature.
- Absence of automatic disease recognition using image processing.
- Inexistence of real-time disease infection detection.
- The absence of highlighting specific areas in the map to identify disease-infected areas, as some diseases, such as bird flu, spread very quickly.
- Lack of comprehensive medical advice.
- Absence of data analysis using real-time data by web scraping.

2 System overview

2.1 Software architecture

In this app, the user should be able to log into the system. They should be able to log into both the website and the intelligent mobile app. The system will use two different databases located on different servers to speed up the speed of the application. To make this happen, two different Application Programming Interface (API) [19] can be used. An

Table 1 Comparison among related products

App name	URL	Unique feature
Poultry App [16]	poultryapp.com	Environment personalization, day & and night time indication
Big Farm Net [17]	bigfarmnet.com	Multiple farms controlling capacity, data analysis in real-time, broiler controlling system
Poultry Manager [18]	manufarms.com	Native language support, business management system, incubator automation

Application Program Interface (API) allows different software applications to communicate and exchange data or services efficiently. One is Google's Firebase [20] API and the other is based on a local Hypertext Pre-processor (PHP) [21] server. The local database server is used as a linked tier to store Internet of Things (IoT) [22] or sensor-related information temporarily, which can be migrated to the main database later. To enable camera real-time infection detection, Google's Tensorflow [23] library is used. The generated Tensorflow model is converted to Tensorflow Light for mobile use. The machine learning algorithm that performs better and faster can be used in the future. However, in this system, a simple Convolutional Neural Networks (CNN) [24] is used. To improve the precision of breed identification and disease classification, advanced ensemble methods [12, 25–28] and Vision Transfomer (ViT) [29–32] can also be applied. Offline Image identification is enabled for breed and disease recognition. A cost calculator is available with an extensive database for poultry farm consultancy. The basic cost calculation is active in offline mode in case of emergency. Medicine data can be added to the database to provide drug suggestions. Food nutrition is vital in terms of creating a poultry food chart, so this module is present in the system. To reduce the database load and at the same time ensure the prevention of data loss, cost estimation reports, food chart reports, and medicine prescriptions will not be saved in the database, rather they will be sent to the user's email address. By this, the data loss possibility is laser because both sides have to delete those data, and at the same time, pressure upon the database will be reduced. A live chat feature should be present in the app. The live chat module should have a different database login so that users can use that module in case of slow internet. In that case, only live chat will be enabled, and other on-line services will be disabled. Some services should also be present on the website for safety. Bird flu notifications will be available both on the website and on the mobile app. The user should be able to mark the location as a bird flu-infected area. After marking the location, the information will be saved to the database, but an immediate change will not occur. Rather, the super admin can manually publish the data from the backend to avoid false signals. Web data scraping [33] should be enabled for statistical analysis. Web scraping is the automated process of extracting data from websites for analysis, research, or application development. The app is optimized for slow Internet connection so that users can handle difficult situations. The temperature and humidity control module works to maintain the proper environment for the poultry farm. Users will be able to control the farm from a distance. The payment system can smoothly manage the donation activity. Figures 1, 2, 3, and 4 represent the overall use case diagram, components diagram, deployment diagram, and database schema diagram for our apps, respectively.

Our system characteristics include:

1. *Centralized AI component*: "GFMC-Intelligent" serves as the central AI-based system, connecting various applications.
2. *Multiple integrated app modules*: the system consists of several interconnected apps, including:
   - Live chat module.
   - Automation module.
   - Donation module.
   - Blog module.
   - Business analysis module.
   - Real-time diseases detection module.
   - Users module.
   - Breed identification module.
   - Disease recognition module.
   - Cost calculator module.
   - Food chart module.
   - Medical support module.
   - Bird flu area map module.
3. *Data access & reports*: each module communicates with "GFMC-Intelligent" via data access modules, providing and receiving data for analysis, reports, and control.
4. *Farm input & camera input*: some applications (e.g., Real-Time Diseases Detection module) require inputs from the farm environment and cameras.
5. *User & farm data integration*: there are clear links for user inputs and farm data feeding into the system for further analysis.
6. *Decision-making support*: the system appears to support decision-making with the help of AI-based predictions and real-time monitoring.
7. *Automation & analytics*: the system emphasizes automation (farm automation, business analysis) and advanced analytics (e.g., real-time machine learning, disease identification).
8. *Communication modules*: user interaction is facilitated through apps like the Live Chat module and Blog module, ensuring engagement and communication.

2.2 Methodologies

The development methodology for the "GFMC-Intelligent" application is a combination of DSDMMIA (Dynamic Systems Development Method for Multiple Integrated Applications) and Scrum, both Agile methodologies designed to ensure flexibility, continuous delivery, and efficient module-based development.

1. Task prioritization (DSDM)

   The project follows DSDM principles for prioritizing tasks and functionalities based on business value and

Fig. 1 Galliformes farm management system use case diagram

necessity. This ensures that critical functionalities—such as the poultry breed and disease identification features—are delivered first, while less essential features are scheduled later.

2. Sprint planning and execution (scrum)

   The project is broken down into sprints, with each sprint focusing on developing and testing specific modules. Sprint cycles typically include:

   i. Sprint planning: setting goals and defining tasks for the next iteration.
   ii. Development: building the selected module (e.g., user module, disease recognition module).
   iii. Testing: conducting unit, integration, and performance testing.
   iv. Feedback and refinement: implementing improvements based on testing feedback before moving to the next sprint.

3. Continuous improvement and feedback loop

   Throughout each sprint, continuous feedback from testing is integrated back into the development cycle. This allows for iterative refinement of modules and features, ensuring the final product is robust and meets user expectations.

   This iterative, flexible approach enables the development team to adapt to changes, integrate new requirements, and continuously improve the application's performance, ensuring both functionality and user experience are optimized. Figure 5shows the system development process of our study.

2.3 Deep learning model development and integration

In our Galliformes Farm Management System, among many modules, some AI-related modules are being introduced to make the application more interactive and user-friendly. An ideal farm management application needs to be able to manage nutrition calculations, provide primary

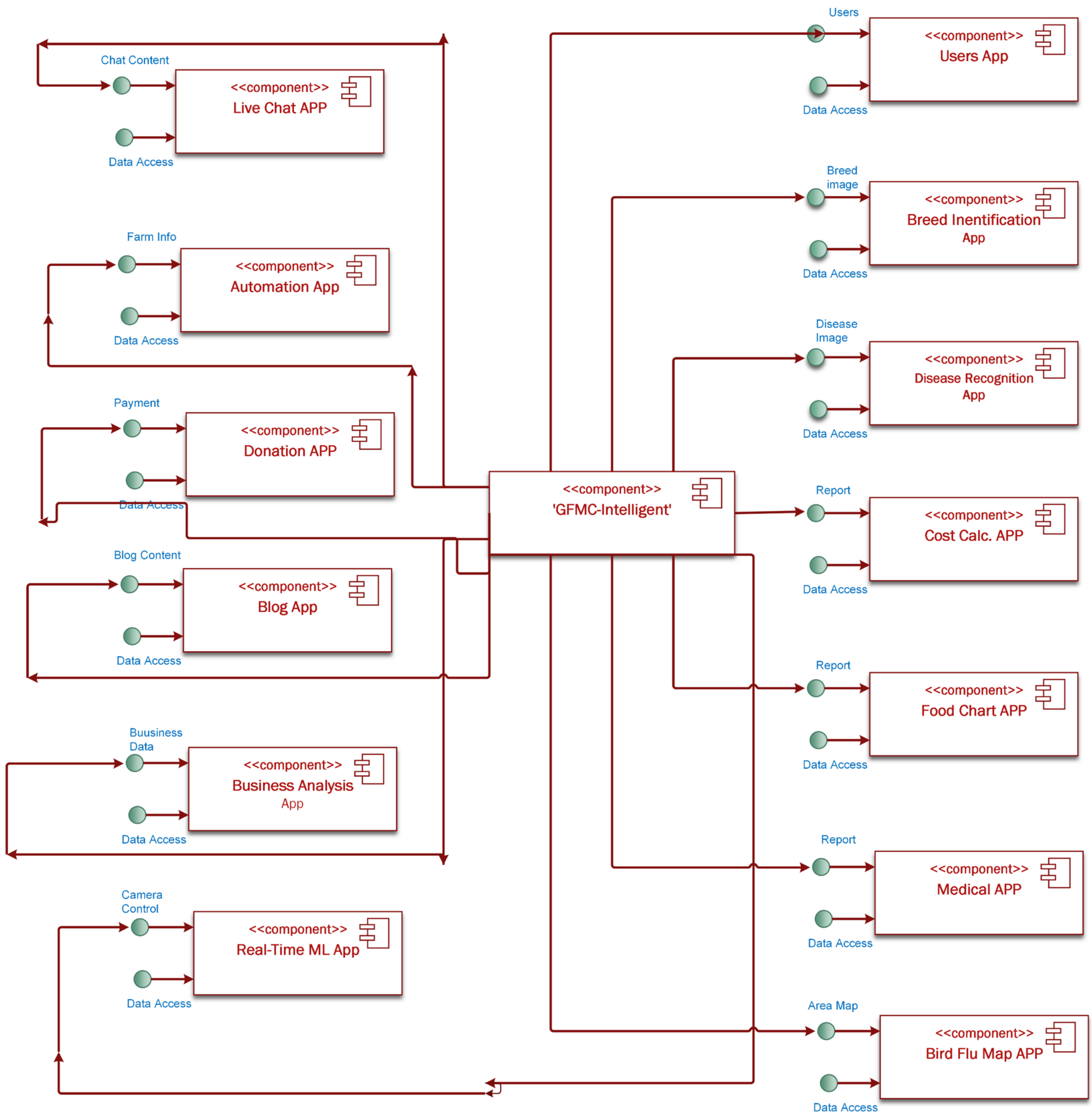

Fig. 2 Diagram of components of the galliformes farm management system

medical information, monitor the farm environment, have the capability of controlling the farm from far away using the Internet, and also be able to interact with the community. However, we have also introduced the concept of breed identification using smartphone-captured images and identifying diseases from fecal images. To do this, we have collected a dataset of fecal images [34] and also developed a dataset of different hen breeds [14] of our own. These datasets were preprocessed and then fed into the machine learning models. We have used different pre-trained models and applied transfer learning to those models, then ensembled those trained AI models using different meta-learners to gain better precision and classification accuracy. After creating the final TensorFlow model, we converted the model to TensorFlowLite version. So, we can use the model in the Android/iOS application. This automatic hen breed and disease detection module is unique in poultry-related smart applications. The overall

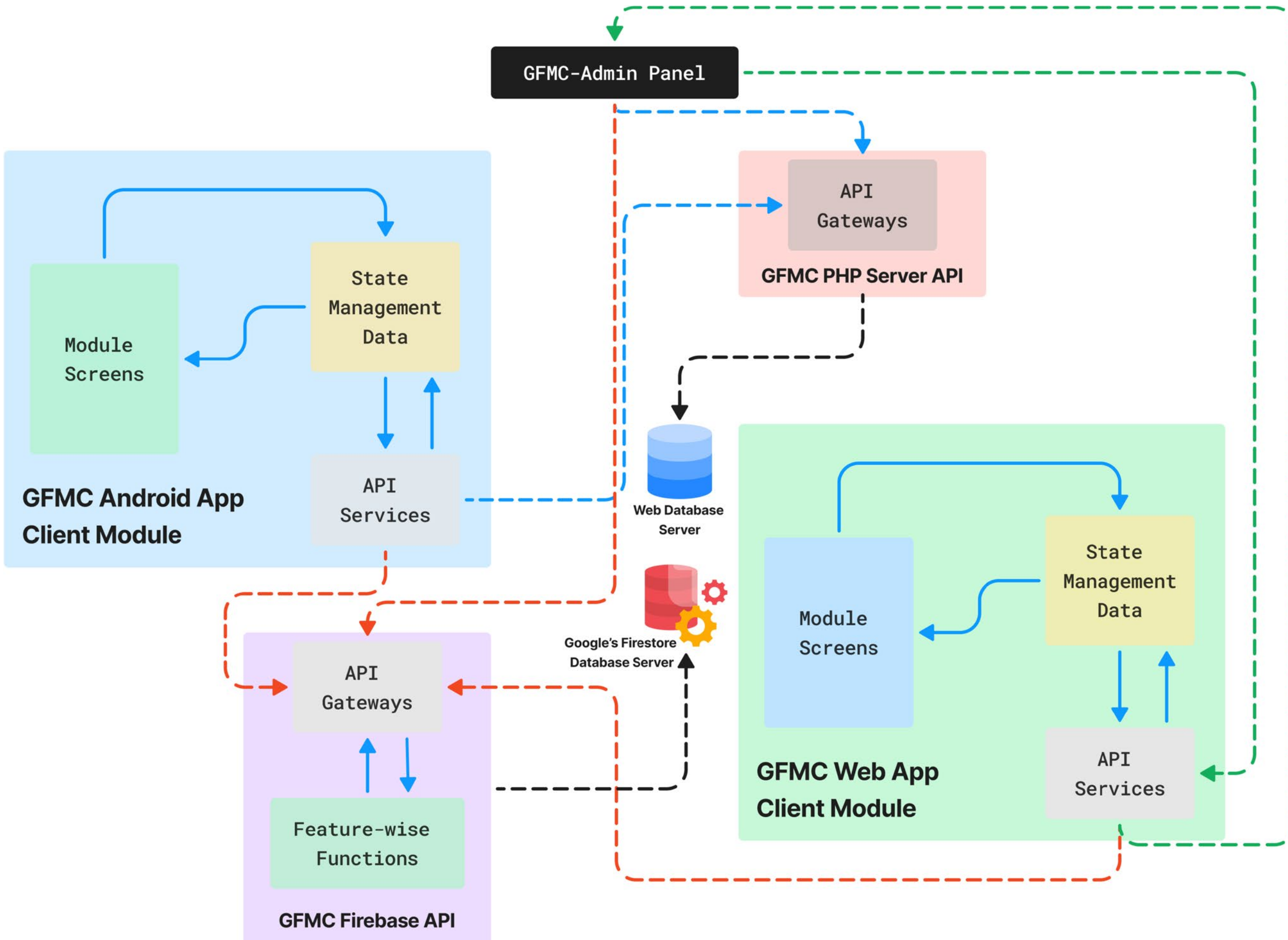

Fig. 3 Galliformes farm management system deployment diagram

framework for developing the ML model is illustrated in Fig. 6. Also, Fig. 7 shows some code snippets of our development.

Transfer learning is a machine learning technique where a model trained on one task is adapted to perform another related task. Instead of starting from scratch, a pre-trained model, usually on large datasets like ImageNet, is fine-tuned to suit a new problem. This approach is highly effective when dealing with limited data, allowing faster convergence and improved accuracy. In our case, we have applied AlexNet [35], MobileNet [36], VGG16, VGG19 [37], and ResNet [38] for image classification part.

2.4 Software functionalities

The main functionalities of our 'Galliformes Farm Management System' app are listed below:

User authentication and authorization: users should be able to log into the system and authenticate and approve themselves.

Management of users: authorized users should be able to control all users on the system. They should be able to restrict and unblock users who attempt to use the system.

Breed identification using machine learning: authorized users should be able to use the image recognition module to identify bird breeds.

Disease recognition using the ML algorithm: users with authority should be able to use the Image Recognition module to identify bird disease.

Food chart generation: authorized users should be able to input the system and generate food charts.

Medicine suggestion module: users with authority should be able to input the system and generate a Medical Consultancy report.

Infected area map: authorized users should be able to report bird flu, which will be approved by the administrator later after authenticating the information.

Also, there are some annex features such as;

- Automated farm controlling module,
- Live-chat module,

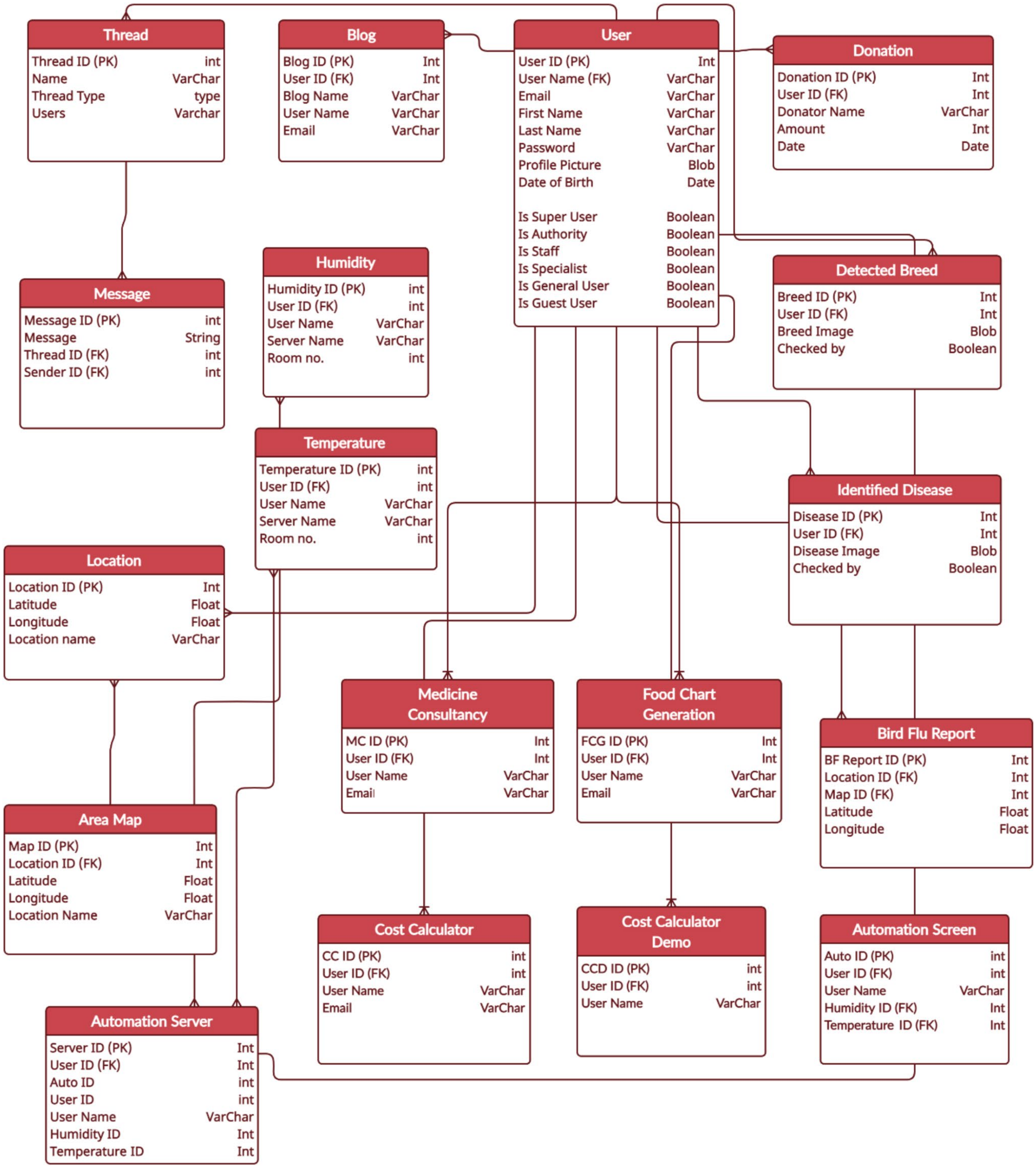

Fig. 4 Galliformes farm management system database schema diagram

- Blog,
- Donation system.

Table 2 represents the metadata of our developed system. The system is available on the GitHub platform under open-source MIT licenses. The listed tools are used in our system to support app development, data science, IoT, and machine learning tasks. Android Studio is an integrated development environment (IDE) for building Android apps, offering features like code editing and debugging.

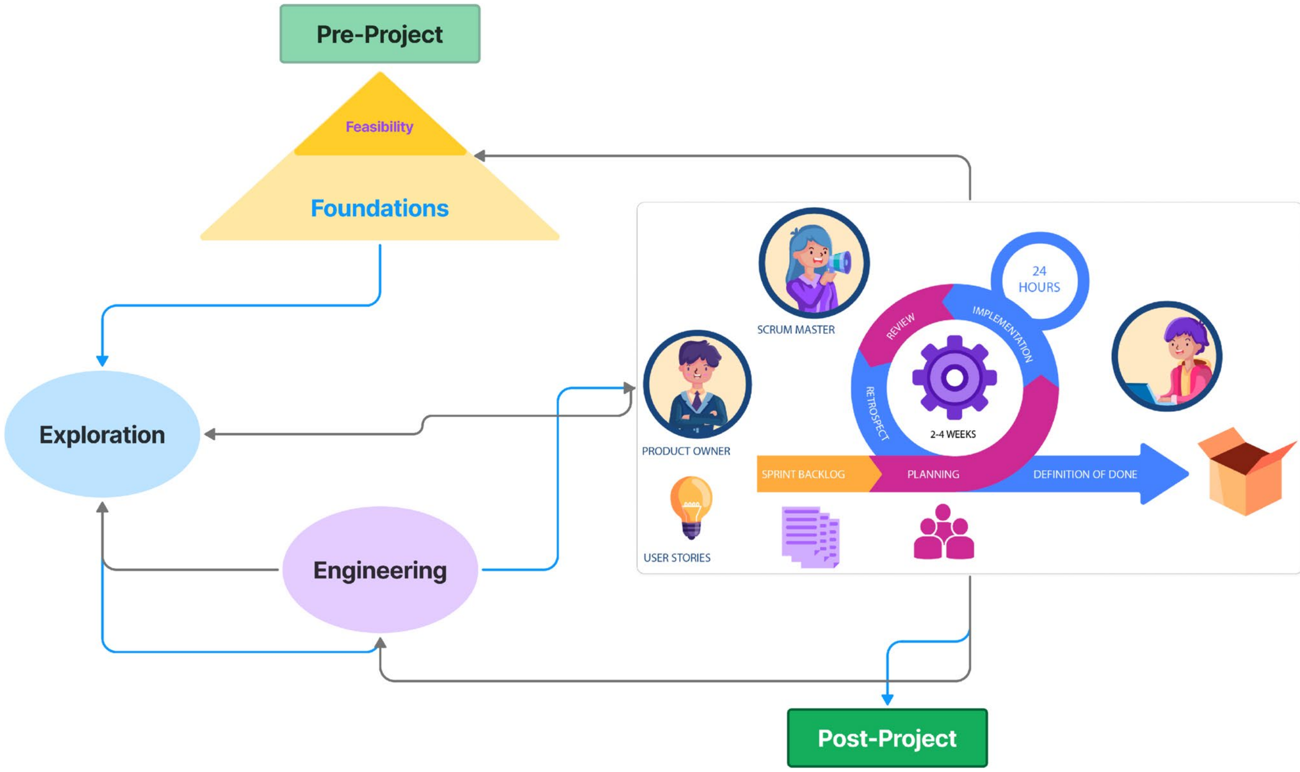

Fig. 5 System development methodology & process

Anaconda Distribution provides a platform for managing Python environments and packages, commonly used in data science and machine learning. Arduino IDE is essential for writing and uploading code to Arduino microcontrollers for IoT and embedded systems projects. Jupyter Lab is used for creating and sharing data analysis workflows and notebooks in Python. SDK 2.7.0 is a software development kit that provides libraries and tools for building specific applications, such as IoT or mobile. Firebase offers cloud-based backend services like real-time databases and user authentication for mobile and web applications. TensorFlow Lite is a lightweight version of TensorFlow, optimized for running machine learning models on mobile and embedded devices, crucial for edge AI and on-device inference.

3 Implementations

The development process of the "GFMC-Intelligent" application follows a structured methodology that combines DSDMMIA (a customized version of the Dynamic Systems Development Method) and Scrum, both of which are Agile methodologies. The DSDM approach prioritizes tasks to ensure critical functionalities are addressed first, while Scrum divides the project into manageable sprints, each functioning as a mini-project for independent development and testing of specific modules. This iterative process allows for continuous improvements and adjustments throughout the development cycle.

The application's programming involves various tools and languages suited to different system components. Flutter, using the Dart language, is chosen for mobile app development due to its cross-platform capabilities, large community support, and ease of integrating libraries. This allows the application to run efficiently on both Android and iOS platforms. For the website, PHP handles backend processes like database management and server-side logic, while JavaScript ensures dynamic client-side behavior. Python is used for developing Machine Learning (ML) models, particularly for real-time image recognition in poultry breed and disease identification. TensorFlow and TensorFlow Lite optimize these models for mobile deployment, ensuring that even devices with lower processing power can handle the operations effectively. R programming is employed in the research phase, where various ML algorithms are tested and compared to select the best fit for image recognition. Additionally, C is used for automation tasks like controlling hardware components (temperature and humidity sensors), leveraging ESP32 microcontrollers to monitor and manage farm environments in real-time.

The system consists of various modules, each presenting unique programming challenges. The User Module

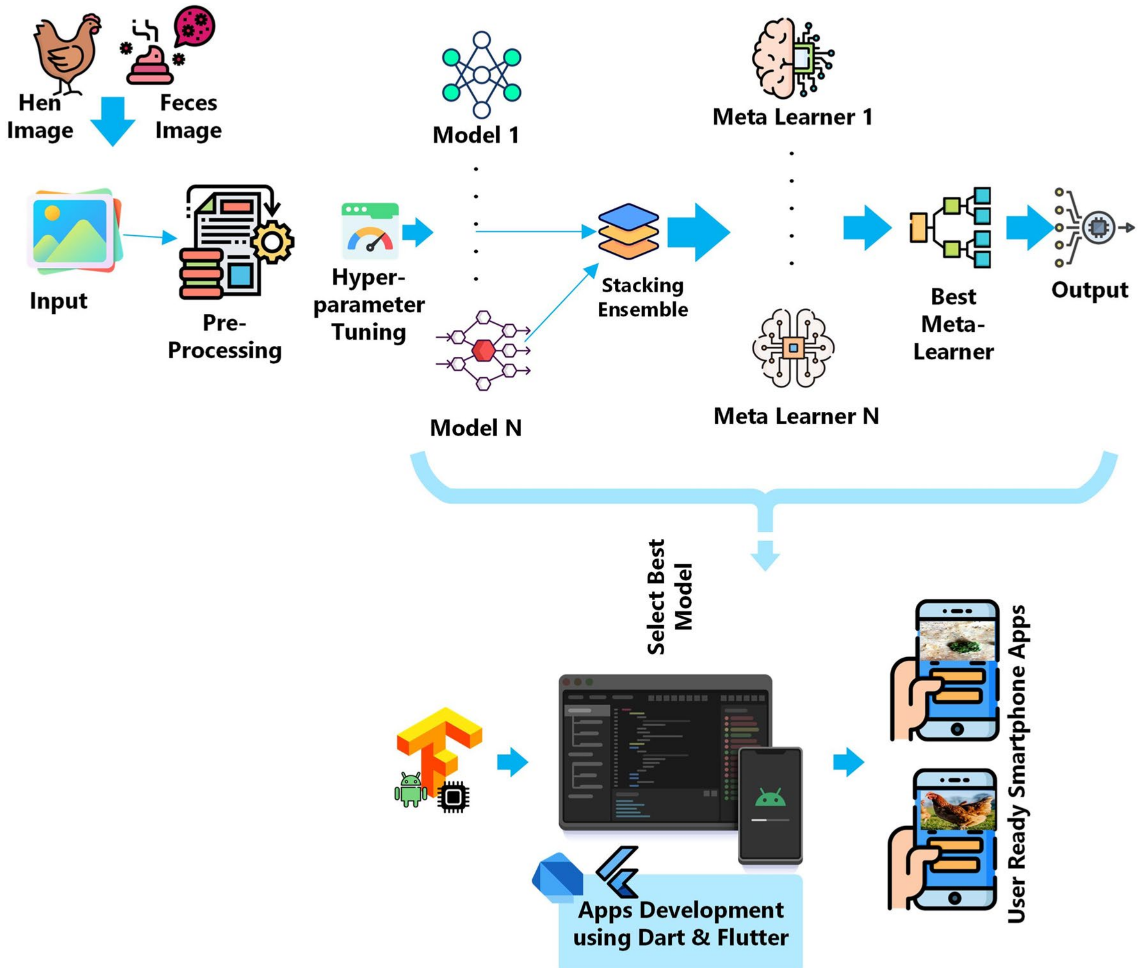

Fig. 6 AI Model development and integration

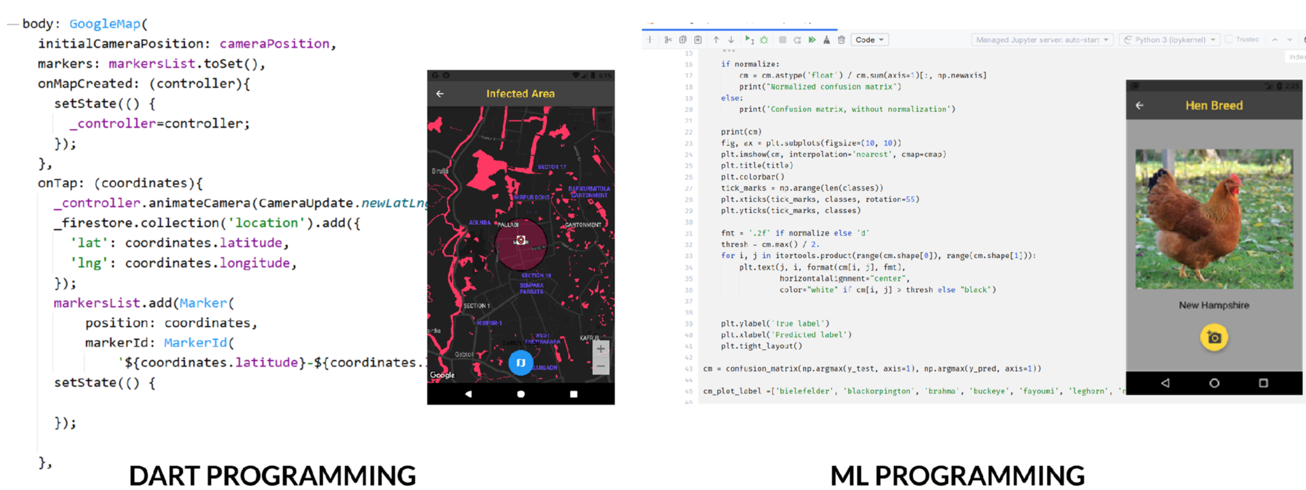

Fig. 7 Example of graphical representation & code snippet for bird flue area annotation (on the left) and hen breed detection (on the right)

Table 2 Metadata of smart galliformes farm management system

Title	Information
Permanent link to code/repository	https://github.com/Galib-Muhammad-Shahriar/GALLIFORMESFARMMANAGEMENT/tree/8568ef2e9f57954a340b2fecfe49aac5cadf834c/1.Mobile_App%20(Dart%20%26Flutter)/TensorFlow_GALLIFORMESFARMMANAGEMENT-master/TensorFlow_GALLIFORMESFARMMANAGEMENT-master
License	MIT License
Software code languages, tools, and services	Dart, Flutter, Python, C/C++
Compilation requirements, operating environments, and dependencies	Android Studio, Anaconda Distribution, Arduino IDE, Jupyter Lab, SDK 2.7.0, firebase, TensorFlow Lite

handles tasks such as registration, login, and profile management via a secure API-based authentication system. The Bird Breed Identification Module uses machine learning and image recognition to identify poultry breeds in real time, functioning both online and offline to accommodate users with inconsistent internet access. Similarly, the Disease Recognition Module utilizes ML models to detect poultry diseases based on image recognition. The Cost Calculation Module accepts user inputs, calculates expenses, and emails reports, avoiding database overload. The Automation Module allows remote monitoring and control of environmental factors such as humidity, temperature, and air pressure, using microcontroller boards like ESP32 and sensors like DHT11. The app also includes a Live Chat and Support Module, enabling users to consult with veterinary experts and support staff in real time.

Development is organized into time boxes, with different phases dedicated to specific tasks. For instance, Iteration 1 focuses on developing the cost calculation module, while later iterations handle disease recognition, automation, and live chat functionalities. This phased approach minimizes errors and facilitates rapid iterations, improving the stability and performance of the application.

High-fidelity prototypes are created for each module, ensuring a clear visual representation of the user interface and interactions. These prototypes help in designing a user-friendly experience, especially for workflows such as cost calculation, bird flu reporting, and real-time infection detection. Wireframes provide practical guidance, ensuring the app is accessible and easy to use for farmers with limited technical expertise.

Testing plays a critical role in the development process, encompassing unit testing (for individual modules), integration testing (to ensure all modules work together), and performance testing (to verify system capacity for real-world loads). Additionally, security testing is conducted to protect the system against potential cyber threats. Each module undergoes rigorous testing within its respective sprint cycle to identify and resolve issues early, ensuring overall system stability.

Deployment follows a structured seven-step process, which includes thorough planning, workflow analysis, and extensive testing. The system is deployed using a dual API-based platform—one API manages user authentication and machine learning tasks, while the other handles general system functions. This setup reduces server load and enhances performance by distributing resource-intensive tasks. The deployment process also integrates Stripe and PayPal for handling payments in the donation module, while Google Maps is used to identify bird flu-infected areas, offering users real-time data on disease outbreaks. The cloud-based infrastructure ensures scalability, enabling the system to support a large number of users.

Practical considerations are vital, especially given that the app is designed for regions with limited internet connectivity. Offline functionality is prioritized, with ML models for breed and disease recognition optimized for mobile use. TensorFlow Lite allows these models to run on mobile devices, making the app responsive and efficient even in low-resource environments. The environmental automation module provides real-time updates every five seconds, enabling remote monitoring and farm control.

The development of GFMC-Intelligent balances cutting-edge technologies like ML with practical considerations for its end-users, such as farmers with limited technical knowledge. The modular, iterative development approach, combined with comprehensive testing and optimization, ensures the app meets the demands of modern poultry farm management efficiently.

4 Illustrative examples

On the app home page, both the online and offline usage options are available. General users can use the interface without logging into the offline version. A dedicated interface was also developed for poultry farmers who can register online and has some extra features. The 'service' interface and the left menu contain all the submodule links such as; Breed Identifications, Disease Recognition, Real-Time

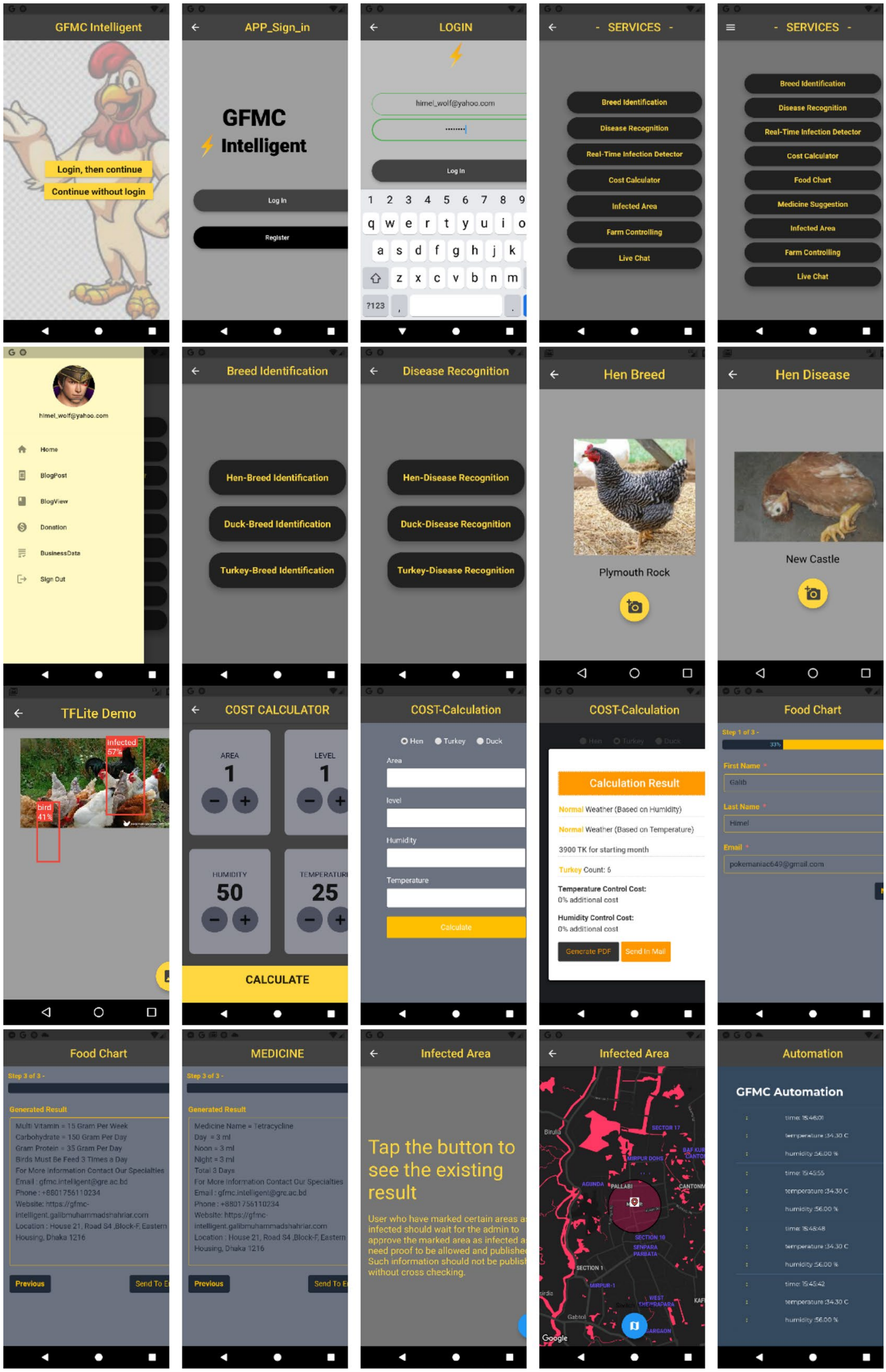

Fig. 8 Interfaces of the applications of various menus and modules of the software

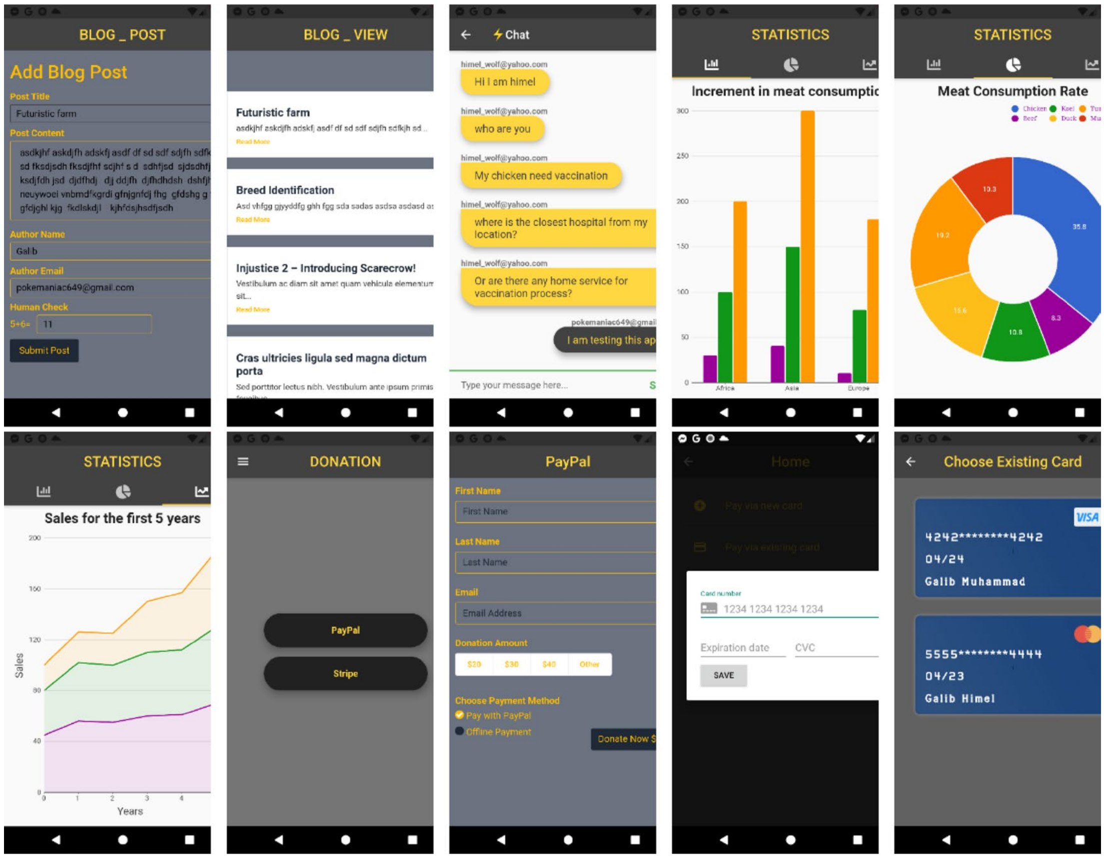

Fig. 8 (continued)

Infection Detector, Cost Calculator, Infected Area, Farm Controlling, Live Chat, Medicine Suggestion, Blog Post, Bloh View, Donation, and Business Data. Figure 8 illustrates the example interface images of this app.

5 Impact

The deployment of our Smart Galliformes Farm Management System ensures a plethora of significant outcomes and values that span economic, social, health, and environmental realms. This intelligent application specifically addresses unemployment challenges in developing nations, simplifying poultry farming for novices and generating job opportunities, especially for local unemployed youths and budget-conscious poultry farmers. By streamlining farm management through reduced on-site surveillance and decreased reliance on physical equipment, this expert system empowers farmers efficiently. Automation reduces the demand for an extensive workforce, making farm management economically viable for novice farmers with limited budgets.

Introducing cutting-edge technologies like machine learning for disease identification and breed categorization, the app fosters agricultural innovation, paving the way for a more sophisticated and efficient farming industry. Alleviating the issue of 'Osmotic Imbalance' [39], the app advocates for poultry farming as a means to increase the supply of animal proteins, thus improving nutritional equilibrium. Offering consulting services, the app supports novice farmers and those lacking agricultural expertise, facilitating skill development and knowledge sharing through live chat modules.

In real-time disease monitoring, the app plays a crucial role in preventing avoidable damage to poultry, directly impacting public health by minimizing the risk of disease transmission between animals and from animals to humans. Addressing concerns related to unsafe poultry consumption, the app encourages the production of safer and more hygienic poultry, mitigating health risks associated with chemical-contaminated meat.

Automation's deployment in farm management not only diminishes the reliance on extensive manual labor, but also contributes to reducing the environmental footprint, aligning with sustainable farming practices by minimizing resource consumption. The integration of machine learning algorithms for the identification of diseases and the categorization of breeds in the app signifies a substantial leap in agricultural technology, significantly contributing to the modernization of farming practices. Also, this kind of application can be implemented in similar domains such as; cattle, sheep, and other livestock animal farming.

6 Conclusions

The Galliformes Farm Management System app, with its AI-based feature, increases its transformative impact on economic, social, health, and environmental dimensions. Using advanced technology, promoting sustainable farming practices, and tackling pressing issues such as unemployment and nutritional imbalances, the app stands out as a comprehensive solution that caters to the dynamic needs of the agricultural sector. With the integration of various modules and features, including cutting-edge AI functionality, the app not only streamlines efficient farm management but also nurtures a sense of community and knowledge-sharing. This positions the app as an invaluable tool that propels the advancement of the agricultural landscape to new heights.

Author contributions Galib Muhammad Shahriar Himel: Conceptualization, Methodology, Machine Learning Model Development, Software Development, Server Development, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing—Original Draft, Visualization. Md. Masudul Islam: Validation, Formal analysis, Resources, Writing—Original Draft, Writing—Original Draft, Visualization. Md. Nasimul Kader: Supervision Mustafizur Rahman: Supervision. Artificial Intelligence-based Smart Galliformes Farm Management System.

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Declarations

Conflict of interest All authors declare that they have no conflict of interest.

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

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